Question ID: e744499e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

An elementary school teacher is ordering x workbooks and y sets of flash cards for a math class. The teacher must order at least 20 items, but the total cost of the order must not be over $80. If the workbooks cost $3 each and the flash cards cost $4 per set, which of the following systems of inequalities models this situation?

Answer

A. $$\begin{aligned} x + y &\geq 20 \\ 3x + 4y &\leq 80 \end{aligned}$$

B. $$\begin{aligned} x + y &\geq 20 \\ 3x + 4y &\geq 80 \end{aligned}$$

C. $$\begin{aligned} 3x + 4y &\leq 20 \\ x + y &\geq 80 \end{aligned}$$

D. $$\begin{aligned} x + y &\leq 20 \\ 3x + 4y &\geq 80 \end{aligned}$$

Correct Answer: A

Rationale

Choice A is correct. The total number of workbooks and sets of flash cards ordered is represented by $x + y$. Since the teacher must order at least 20 items, it must be true that $x + y \geq 20$. Each workbook costs $3; therefore, $3x$ represents the cost, in dollars, of x workbooks. Each set of flashcards costs $4; therefore, $4y$ represents the cost, in dollars, of y sets of flashcards. It follows that the total cost for x workbooks and y sets of flashcards is $3x + 4y$. Since the total cost of the order must not be over $80, it must also be true that $3x + 4y \leq 80$. Of the choices given, these inequalities are shown only in choice A.

Choice B is incorrect. The second inequality says that the total cost must be greater, not less than or equal to $80. Choice C incorrectly limits the cost by the minimum number of items and the number of items with the maximum cost. Choice D is incorrect. The first inequality incorrectly says that at most 20 items must be ordered, and the second inequality says that the total cost of the order must be at least, not at most, $80.